RetailVision AI: A Real-Time Intelligent Video Analytics for Retail Customer Behavior Analysis Using YOLOv8 and DeepSORT

Naveen Gundyagol
Department of Computer Science and Engineering (Artificial Intelligence & Machine Learning)
Joy University
Internship Organization: TANSAM (Tamil Nadu Smart and Advanced Manufacturing Centre)
Roll No.: 2023BTIA021

Abstract**—***Retail surveillance systems generate substantial volumes of video data that are conventionally used only for security monitoring, leaving valuable customer behavior information largely unexplored. This paper presents RetailVision AI, an artificial-intelligence-based system that converts ordinary retail surveillance footage into actionable business intelligence. The proposed pipeline integrates YOLOv8 for real-time person detection, DeepSORT for multi-object tracking, and OpenCV for video processing in order to automatically detect customers, assign persistent identities, and monitor their movement across a store. The system computes zone-wise visit counts, estimates customer dwell time within predefined regions, and generates movement heatmaps that reveal high-traffic areas. All analytical outputs—including key performance indicators, the processed video, and CSV reports—are presented through an interactive Streamlit dashboard. Results obtained on recorded retail video demonstrate reliable customer counting, consistent identity retention under brief occlusion, and meaningful zone and dwell-time analytics, validating the applicability of the system for data-driven retail operations and store layout optimization.*

Index Terms**—***Computer Vision, Object Detection, YOLOv8, DeepSORT, Multi-Object Tracking, Retail Analytics, Dwell Time Estimation, Heatmap Generation, Streamlit Dashboard.*

I. INTRODUCTION

The retail industry increasingly relies on data-driven decision-making to improve customer experience, optimize store layouts, and enhance operational efficiency. While modern retail stores are equipped with surveillance cameras, the recorded video is primarily used for security purposes, leaving customer behavior information largely unexamined.

Recent advancements in artificial intelligence (AI) and computer vision have enabled surveillance footage to be analyzed automatically, providing meaningful insight into customer movement, shopping patterns, and store occupancy. Object detection and multi-object tracking allow businesses to understand how customers interact with different areas of a store without requiring manual observation.

This work presents RetailVision AI, an AI-powered retail video analytics system capable of detecting customers, tracking their movement across video frames, estimating dwell time within predefined zones, and generating customer movement heatmaps. The processed information is presented through an interactive Streamlit dashboard, enabling users to visualize key retail analytics in a simple and intuitive manner. The system integrates YOLOv8 for real-time person detection, DeepSORT for multi-object tracking, OpenCV for video processing, and Streamlit for dashboard visualization, together transforming ordinary retail surveillance footage into actionable business intelligence.

Manually monitoring customer movement, estimating dwell time, and identifying high-traffic areas is time-consuming, inefficient, and prone to human error. The objective of this work is therefore to develop an AI-powered retail video analytics system capable of automatically detecting customers, tracking their movement, analyzing customer activity within predefined store zones, and presenting meaningful business insights through an interactive dashboard. The specific objectives are:

1) Detect customers from retail surveillance videos using YOLOv8.
2) Perform multi-object tracking using the DeepSORT algorithm.
3) Calculate the number of unique customers and peak store occupancy.
4) Analyze customer visits across predefined store zones.
5) Estimate customer dwell time within each zone.
6) Generate customer movement heatmaps for visual analysis.
7) Develop an interactive Streamlit dashboard for presenting retail analytics.

The work focuses on analyzing recorded retail surveillance videos to generate customer behavior analytics, performing person detection, customer tracking, zone-wise analytics, dwell time estimation, and heatmap generation using computer vision techniques. Although the current implementation processes recorded videos, the overall pipeline can be extended to support live CCTV feeds, cloud deployment, and advanced retail analytics in future implementations.

II. RELATED WORK

Retail video analytics refers to the process of extracting meaningful business information from surveillance footage using computer vision and artificial intelligence. Instead of using CCTV cameras solely for security, retailers can analyze customer movement, occupancy, dwell time, and shopping behavior to improve store operations and customer experience, identify high-traffic areas, evaluate customer engagement, and optimize product placement and workforce planning.

YOLO (You Only Look Once) is a real-time object detection algorithm designed to identify objects accurately while maintaining high processing speed [1]. YOLOv8, developed by Ultralytics, improves detection accuracy and efficiency relative to earlier versions, making it suitable for real-time applications [5]. In this work, YOLOv8 is used to detect customers in each video frame, and the resulting bounding boxes serve as input to the tracking module.

DeepSORT (Deep Simple Online and Realtime Tracking) is a multi-object tracking algorithm that assigns unique identities to detected objects and maintains those identities across consecutive video frames [2], [3]. Unlike simple tracking methods, DeepSORT combines motion prediction with appearance information, allowing customers to retain consistent identities even under temporary occlusion. In this work, DeepSORT is responsible for assigning unique customer IDs, preventing duplicate counting, and enabling dwell time estimation.

OpenCV [4] is used for reading video frames, drawing bounding boxes, displaying tracking information, defining analytical zones, generating heatmaps, and saving processed video output. Streamlit [6] is an open-source Python framework used to build the interactive dashboard with minimal code, while Pandas [7] supports the underlying data processing for analytics generation.

III. PROPOSED SYSTEM

A. System Workflow

The proposed system reads the retail surveillance video and extracts frames using OpenCV. Customers are detected in each frame using the YOLOv8 model, and unique identities are assigned using DeepSORT. Customer movement is tracked across consecutive frames, visits within predefined store zones are identified, and dwell time is calculated for each zone. A customer movement heatmap is generated, analytical results are stored in CSV files, and all analytics are displayed through the Streamlit dashboard.

B. System Architecture

Retail Video Input
↓
Video Frame Extraction (OpenCV)
↓
YOLOv8 Person Detection
↓
DeepSORT Multi-Object Tracking
↓
Customer Counting | Zone Analytics | Dwell Time
↓
Heatmap Generation
↓
CSV Report Generation
↓
Streamlit Dashboard

Fig. 1. System architecture of the proposed RetailVision AI pipeline.

C. System Modules

1) Video Processing Module: Reads the retail surveillance video using OpenCV and extracts individual frames for further analysis; also handles video writing after tracking annotations are added.

2) Customer Detection Module: The YOLOv8 model detects customers in every frame by identifying the “person” class and generates bounding boxes that serve as input to the tracking module.

3) Customer Tracking Module: DeepSORT assigns a unique tracking ID to every detected customer, with IDs remaining consistent across frames to enable accurate counting and movement analysis.

4) Zone Analytics Module: Monitors predefined regions inside the store to determine customer visits; each time a tracked customer enters a zone, the corresponding visit count is updated.

5) Dwell Time Analysis Module: Calculates the duration each customer spends inside a predefined zone, helping to estimate customer engagement with different areas of the store.

6) Heatmap Generation Module: Accumulates customer movement coordinates collected during tracking to generate a heatmap that visually represents customer activity density.

7) Dashboard Module: Displays processed video, key performance indicators, heatmaps, zone analytics, and dwell time reports through an interactive Streamlit interface.

D. Advantages

1) Automated customer detection and tracking.
2) Accurate customer counting without manual observation.
3) Real-time, zone-wise customer analytics.
4) Customer dwell time estimation.
5) Interactive dashboard for business visualization.
6) Scalable design extensible to live CCTV deployment.

IV. IMPLEMENTATION

The system was developed in Python and integrates multiple computer vision and data processing libraries to analyze retail surveillance footage. The implementation comprises customer detection, multi-object tracking, zone analytics, dwell time estimation, heatmap generation, and dashboard visualization.

A. Development Environment

Component	Specification
Programming Language	Python
IDE	Visual Studio Code
Dashboard Framework	Streamlit
Computer Vision Library	OpenCV
Object Detection	YOLOv8
Object Tracking	DeepSORT

Component	Specification
Data Processing	Pandas
Operating System	Windows 11 (64-bit)

B. Video Processing

The input retail surveillance video is processed frame by frame using OpenCV. Each frame is resized and passed to the object detection model for identifying customers. Processed frames are annotated with bounding boxes, tracking IDs, and zone boundaries before being written to the output video.

C. Customer Detection

YOLOv8 detects customers in each video frame, considering only the person class. The detection module generates bounding boxes with associated location, coordinates, and confidence, which are passed to the tracking module.

D. Multi-Object Tracking

DeepSORT assigns a unique identity to each detected customer, with the assigned ID remaining consistent across consecutive frames. This information is used to count unique customers, estimate peak occupancy, monitor movement, and prevent duplicate counting.

E. Zone Analytics

The retail store is divided into predefined analytical zones. Whenever a tracked customer enters a zone, the corresponding visit counter is updated, allowing high-traffic locations to be identified. The resulting zone analytics are exported to a CSV file and displayed on the dashboard.

F. Dwell Time Estimation

The system records the entry and exit timestamps of a customer within a predefined zone, and the difference between these timestamps is used to calculate dwell time, which helps estimate customer engagement within different store sections.

G. Heatmap Generation

During tracking, the center point of each detected customer is recorded. These coordinates are accumulated over the entire video to generate a movement heatmap that highlights areas of high customer activity.

H. Streamlit Dashboard

A Streamlit dashboard visualizes the generated analytics, displaying the processed retail video, key performance indicators, the customer movement heatmap, zone analytics, and dwell time analysis through an interface that does not require technical knowledge to interpret.

I. Output Files

File	Description
retail_analytics_output.mp4	Processed video with tracking annotations
retail_heatmap_overlay.png	Customer movement heatmap
summary.csv	Summary metrics
zone_analytics.csv	Zone-wise visit statistics
dwell_time.csv	Customer dwell time information

V. RESULTS AND DISCUSSION

The developed application successfully analyzes retail surveillance video by detecting customers, tracking their movement, generating analytical insights, and displaying the results through an interactive dashboard.

A. Customer Detection and Tracking

The YOLOv8 model successfully detected customers in the retail surveillance video, and DeepSORT assigned unique IDs to each detected customer, enabling continuous tracking across multiple frames. This approach ensured accurate customer counting while minimizing duplicate detections.

B. Dashboard Visualization

The Streamlit dashboard provides a user-friendly interface for visualizing the generated retail analytics, displaying processed video output, key performance indicators, customer movement heatmaps, zone analytics, and dwell time information within a single platform.

C. Customer Movement Heatmap

The generated heatmap represents customer movement patterns throughout the store. Areas with higher customer activity appear with greater intensity, allowing identification of frequently visited regions and overall traffic distribution.

D. Zone Analytics and Dwell Time

The predefined analytical zones successfully recorded customer visits during processing, helping identify high- and low-traffic areas for store layout optimization. The system also successfully calculated dwell time within predefined zones, providing insight into customer engagement with different store sections.

E. Discussion

By integrating customer detection, tracking, zone analysis, dwell time estimation, and heatmap generation, the system converts surveillance footage into meaningful business insights. The interactive dashboard further improves usability by presenting all analytical information in a structured and accessible manner. The results validate the practical applicability of the system for retail customer behavior analysis and operational decision-making.

VI. CONCLUSION AND FUTURE SCOPE

RetailVision AI demonstrates the practical application of artificial intelligence and computer vision in retail analytics. The developed system successfully detects customers, tracks their movement, analyzes customer activity within predefined store zones, estimates dwell time, and generates customer movement heatmaps from retail surveillance video. The integration of YOLOv8, DeepSORT, OpenCV, and Streamlit resulted in a complete end-to-end retail analytics solution that enables users to understand customer behavior, identify high-traffic areas, and obtain meaningful business insights without manual observation.

Although the system achieves its intended objectives, several enhancements can be incorporated in future versions:

1) Support real-time analysis using live CCTV camera feeds.
2) Integrate cloud-based storage and remote dashboard access.
3) Implement customer demographic analysis such as age and gender estimation.
4) Develop customer trajectory and shopping path analysis.
5) Generate automated business reports and trend analysis.
6) Extend the system to support multiple camera feeds simultaneously.

The modular design of the proposed system makes these enhancements feasible while maintaining the existing architecture.

REFERENCES

[1] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You Only Look Once: Unified, Real-Time Object Detection," in Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR), 2016.

[2] A. Bewley, Z. Ge, L. Ott, F. Ramos, and B. Upcroft, "Simple Online and Realtime Tracking," in Proc. IEEE Int. Conf. Image Processing (ICIP), 2016.

[3] N. Wojke, A. Bewley, and D. Paulus, "Simple Online and Realtime Tracking with a Deep Association Metric," in Proc. IEEE Int. Conf. Image Processing (ICIP), 2017.

[4] OpenCV Development Team, OpenCV Documentation. [Online]. Available: https://opencv.org

[5] Ultralytics, YOLOv8 Documentation. [Online]. Available: https://docs.ultralytics.com

[6] Streamlit Inc., Streamlit Documentation. [Online]. Available: https://docs.streamlit.io

[7] Pandas Development Team, Pandas Documentation. [Online]. Available: https://pandas.pydata.org

[8] Python Software Foundation, Python Documentation. [Online]. Available: https://docs.python.org/3